Abbreviations

~M_4 = signr(preQuoter(mRPE_Good,hbind(pkisk2_1),hpol(
polP3_Members3,OnlyGood,OnlyGood,OnlyGood,Req
exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,
Assign(P3)),Req(exact(mE2,exact(s2,exact(prodA),
V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1)

~M_5 = pkisk2_1

~M_6 = hpol(polP2_Members3,OnlyGood,OnlyGood,OnlyGood,
Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,
Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3)))))

~M_7 = signr(preQuoter(mRPE_Good,hbind(pkisk2_1),hpol(
polP2_Members3,OnlyGood,OnlyGood,OnlyGood,Req(
exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,
Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1)

~M_8 = pkisk2_1

~M_9 = hpol(polP1_Members3,OnlyGood,OnlyGood,OnlyGood,
Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,
Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3)))))

~X_1 = (P2, ~M_9,hpol(polP1_Members3,OnlyGood,OnlyGood,
OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),
V1,OnlyGood,Assign(P3)),Req(exact(mE2,exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(
mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(
P3))))))
= (P2,signr(preQuoter(mRPE,Good,hbind(pkisk2_1),hpol(polP2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pkisk2_1),hpol(polP1_Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2,exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pkisk2_1),hpol(polP3_Members3,Both,Both,Both,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))

~M_10 = signr(preQuoter(mRPE,Good,hbind(pkisk2_2),hpol(
polP3_Members3,Both,Both,Both,Req(exact(mE11,
exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),
Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,
Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),
V1,OnlyGood,Assign(P3)))))),akRPE_1)

~M_11 = pkisk2_2

~M_12 = hpol(polP3_Members3,Both,Both,Both,Req(exact(
mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(
P1),Req(exact(mE2),exact(s2),exact(prodA),V1,
OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3))))

~X_2 = (P3, ~M_10, ~M_11,hpol(polP3_Members3,Both,Both,
Both,Req(exact(mE11,exact(s11,exact(prodA),V1,
OnlyGood,Assign(P3)),Req(exact(mE2,exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(
mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(
P3))))))
= (P3,signr(preQuoter(mRPE,Good,hbind(pkisk2_2),hpol(polP3,Members3,Both,Both,Both,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pkisk2_1),hpol(polP3,Members3,Both,Both,Both,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),ak1_1)

~M_13 = signr(policyMsg(pkisk2_1),hpol(P3,Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3)))))),ak1_1)

~M_14 = signr(policyMsg(pkisk2_1),hpol(P1_Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3)))))),ak1_1)

~X_3 = (P1, ~M_13, ~M_14,hpol(polP1_Members3,Both,Both,
Both,Req(exact(mE11,exact(s11,exact(prodA),V1,
OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(
mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(
P3))))))
= (P1,signr(preQuoter(mRPE,Good,hbind(pkisk2_1),hpol(polP1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pkisk2_2),hpol(polP3_Members3,Both,Both,Both,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),ak1_1)

~M_15 = signr(policyMsg(pkisk2_2),hpol(P2_Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3)))))),ak2_1)

~M_16 = signr(policyMsg(pkisk2_1),hpol(P2_Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3)))))),ak2_1)

~X_5 = (P1, ~M_7, ~M_8,hpol(polP1_Members3,OnlyGood,OnlyGood,
OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),
V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(
mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(
P3))))))
= (P1,signr(preQuoter(mRPE,Good,hbind(pkisk1_1),hpol(polP1_Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pkisk1_1),hpol(polP1_Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),ak1_1)

~X_6 = (P2, ~M_4, ~M_5,hpol(P2_Members3,OnlyGood,OnlyGood,
OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),
V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(
mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(
P3))))))
= (P2,signr(preQuoter(mRPE,Good,hbind(pkisk2_1),hpol(polP2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pkisk2_2),hpol(polP3_Members3,Both,Both,Both,Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),ak1_1)

~M_17 = signr(policyMsg(pkisk1_1),hpol(P3,Members3,
Both,Both,Both,Req(exact(mE11),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),
Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,
Assign(P3))),ak1_2)

~M_18 = signr(policyMsg(pkisk2_1),hpol(P3,Members3,
Both,Both,Both,Req(exact(mE11),exact(s1),exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),
Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,
Assign(P3))),ak1_2)

Honest Process

Attacker

{1} new akADV_1

{2} new akTE1_1

{3} new akADV_1

(~M_M_1,~M_2,~M_3) = (pkisk4RPE_1),pkisk4TE1_1),
pkisk4ADV_1),skADV_1)

Beginning of process rpe1

{9} new sk1_1

{11} event PolicyDeclared(P1,pol(P1_Members3,OnlyGood,
OnlyGood,OnlyGood,Req(exact(mE11,exact(s11,exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),
Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,
Assign(P3))))

Beginning of process rpe2

{5} new sk2_1

{72} event PolicyDeclared(P2,pol(P2_Members3,OnlyGood,
OnlyGood,OnlyGood,Req(exact(mE11,exact(s11,exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),
Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,
Assign(P3))))

Beginning of process rpe3

(P2, ~M_4, ~M_5, ~M_6)

(P1, ~M_7, ~M_8, ~M_9)

~X_1

{20} event EvidenceChecked(P1,P2,pkisk2_1),hpol(
pol(P2_Members3,OnlyGood,OnlyGood,OnlyGood,Req(
exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3))))))

PermissiveRpeTCB

{132} new sk3_2

{134} event PolicyDeclared(P1,pol(P3_Members3,Both,
Both,Both,Req(exact(mE11),exact(s1),exact(prodA),
V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(
mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(
P3))))

(P3, ~M_10, ~M_11, ~M_12)

~X_2

{27} event EvidenceChecked(P3,P3,pkisk3_2),hpol(
pol(P3_Members3,Both,Both,Both,Req(exact(mE11),
exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),
Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,
Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),
V1,OnlyGood,Assign(P3))))

{28} event QuotesComplete(P1)

{29} event PolicySent(P1,pkisk2_1),pol(P1_Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3))))

(P1,P2,~M_13)

{31} event PolicySent(P1,pkisk3_2),pol(P1_Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3))))

(P1,P3,~M_14)

~X_3

{81} event EvidenceChecked(P2,P1,pkisk3_2),hpol(
pol(P3_Members3,Both,Both,Both,Req(exact(mE11,
exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),
Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,
Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),
V1,OnlyGood,Assign(P3))))))

~X_4

{88} event EvidenceChecked(P2,P3,pkisk1_1),hpol(
pol(P1_Members3,OnlyGood,OnlyGood,OnlyGood,Req(
exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3))))))

{89} event QuotesComplete(P2)

{90} event PolicySent(P2,pkisk3_2),pol(P2_Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3))))

(P2,P1,~M_15)

{92} event PolicySent(P2,pkisk1_1),pol(P2_Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3))))

(P2,P3,~M_16)

(P2,P1,~M_16)

{35} event PolicyAuthenticated(P1,P2,pol(P2_Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3))))))

{38} event PolicyAccepted(P1,P2,pol(P2_Members3,
OnlyGood,OnlyGood,OnlyGood,Req(exact(mE11,exact(
s11,exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(
mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(
P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,
OnlyGood,Assign(P3))))))

{39} event PolicyAcceptedKeys(P1,P2,pkisk1_1),pk(
sk2_1),pol(P2_Members3,OnlyGood,OnlyGood,OnlyGood,
Req(exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3))))

~X_5

{143} event EvidenceChecked(P3,P1,pkisk1_1),hpol(
pol(P1_Members3,OnlyGood,OnlyGood,OnlyGood,Req(
exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3))))))

~X_6

{150} event EvidenceChecked(P3,P2,pkisk2_1),hpol(
pol(P2_Members3,OnlyGood,OnlyGood,OnlyGood,Req(
exact(mE11,exact(s11,exact(prodA),V1,OnlyGood,
Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),
V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3))))))

{151} event QuotesComplete(P3)

{152} event PolicySent(P3,pkisk1_1),pol(P3_Members3,
Both,Both,Both,Req(exact(mE11,exact(s11,exact(
prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),
Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,
Assign(P3))))

(P3,P1,~M_17)

{154} event PolicySent(P3,pkisk2_1),pol(P3_Members3,
Both,Both,Both,Req(exact(mE11,exact(s11,exact(
prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),
Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,
Assign(P3))))

(P3,P2,~M_18)

(P3,P1,~M_17)

{42} event PolicyAuthenticated(P1,P3,pol(P3_Members3,
Both,Both,Both,Req(exact(mE11,exact(s11,exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),
Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,
Assign(P3))))))

{45} event PolicyAccepted(P1,P3,pol(P3_Members3,
Both,Both,Both,Req(exact(mE11,exact(s11,exact(
prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),
exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),
Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,
Assign(P3))))))

{46} event PolicyAcceptedKeys(P1,P3,pkisk1_1),pk(
sk3_2),pol(P3_Members3,Both,Both,Both,Req(exact(
mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(
P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,
OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),
exact(prodA),V1,OnlyGood,Assign(P3))))

{47} event PoliciesComplete(P1)

{50} event FullJoin(P1,mkstart(OnlyGood,OnlyGood,
OnlyGood,Req(exact(mE11,exact(s11,exact(prodA),
V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),
exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(
mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(
P3)))))

A trace has been found.